Electronic AC / DC Load

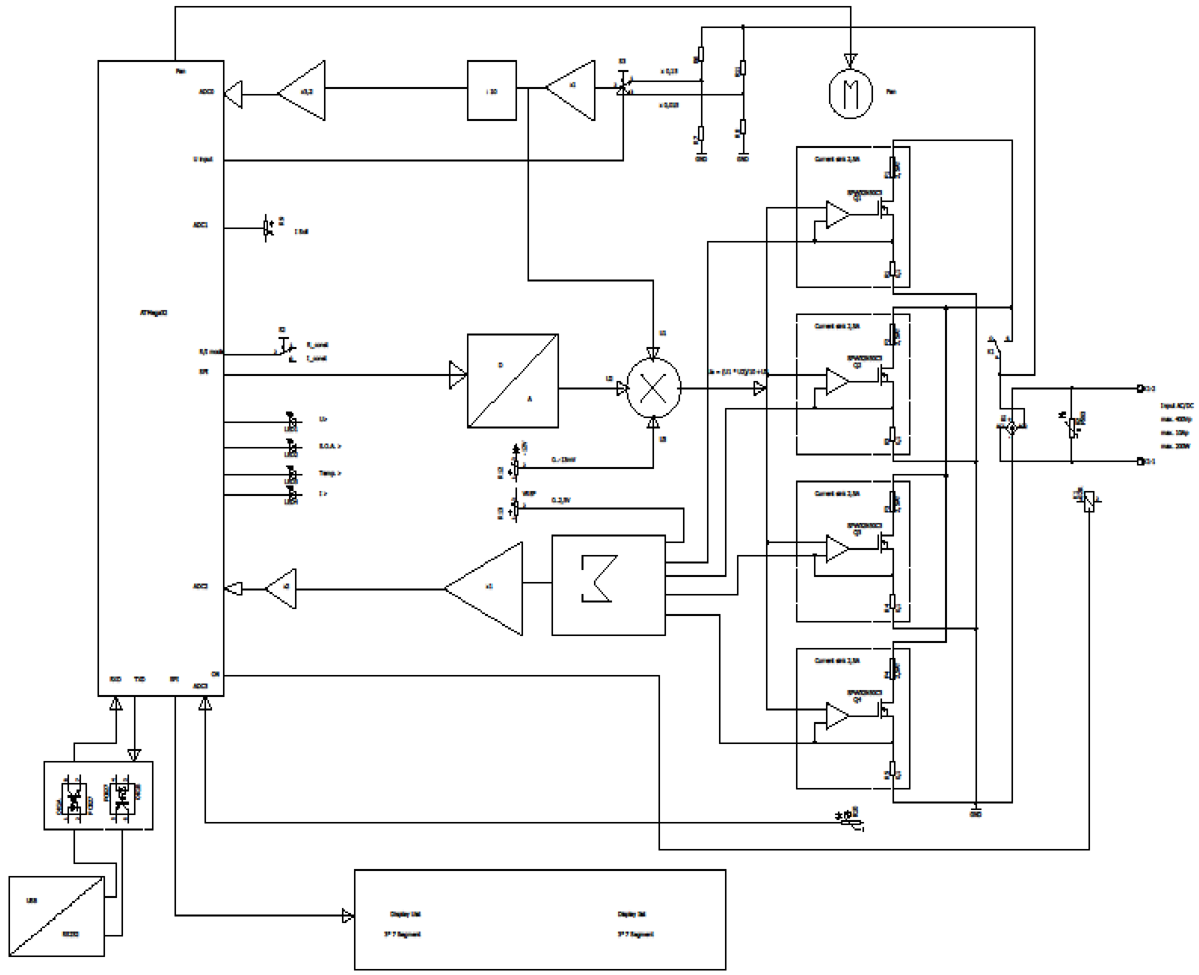
Description

1. General

This electronic load is suitable for DC and AC loads up to 400V (peak) and 10A (peak). The maximum power dissipation is 200W.Therefore, this electronic load can also be used for mains related circuits. It also can be remote controlled via isolated USB interface.

2. Principle Operation

The principle of this electronic load is described here:



3. Description of the hardware components

The hardware is located on a single PCB. Here schematics and layout of the electronic load:

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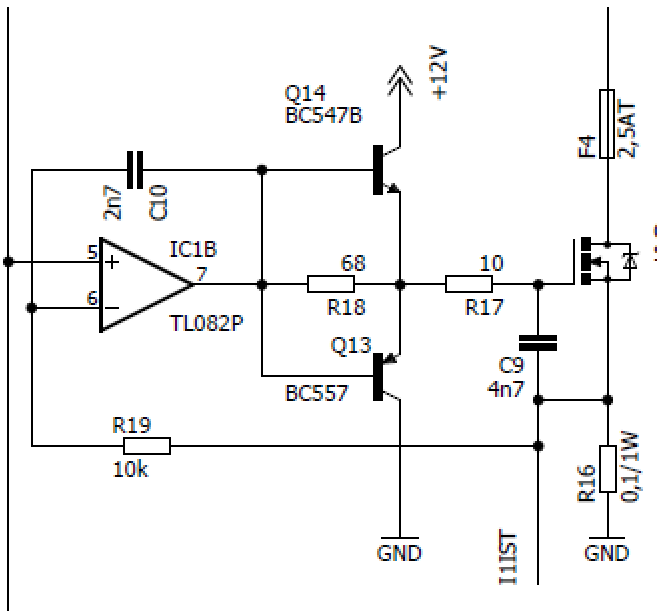
3.1 Constant current sink

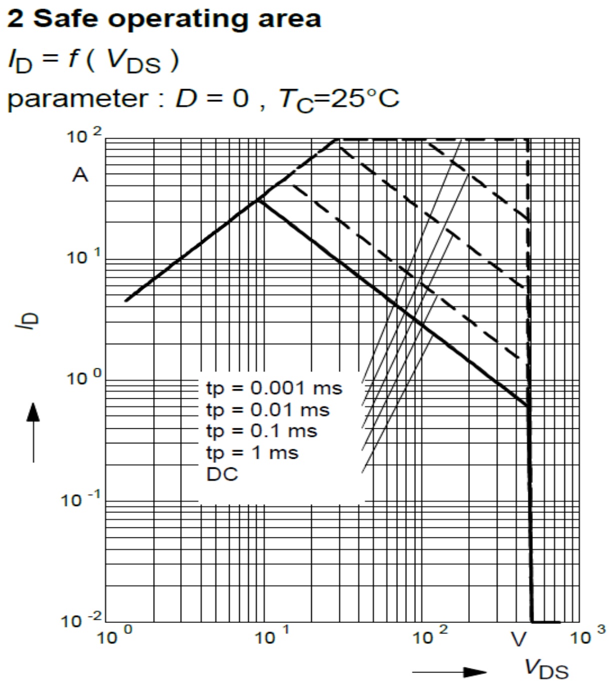
As a load resistor, a „classical“ constant current sink will be used:



The control voltage is connected to the non inverting input of the Opamp. The resulting load current is equal to Control voltage/R16. R19 and C10 prevents the circuitry from oscillating.

For realizing a load current of 10A, this circuitry is paralleled four times.

The current through each unit is therefore 2,5A. The requrirement for the electronic load is a peak voltage of 400V, therefore a Mosfet type SPW32N50C3 is used. A look into the SOA curve shows:



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The maximum current for DC operation is 0,7A @400V. For four units this makes 2,8A, which is 1,12kW!!!!

But thats not possible because of the thermal limitations of the heatsink.

The used heatsink is LK75 from ELV Elektronik which operates with a Papst fan @12V.

Datasheet:

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With this combination, the heatsink has a thermal resistance of 0,44K/W. The thermal resistance of the isolating foile (1,25K/W) and the Junction to case of the Transistor (also 0,44K/W) must be added. Therefore the total thermal resistance is 2,13 K/W.

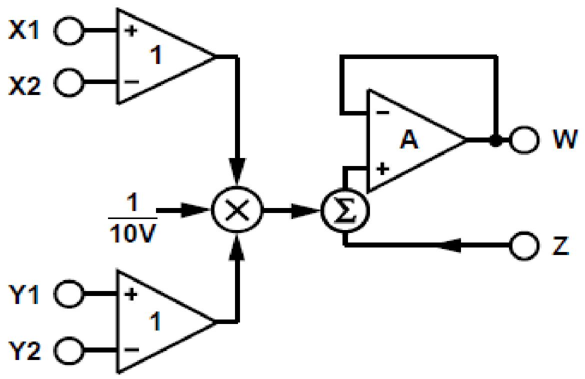
The resulting maximum power dissipation of a transistor (@150°C maximum Junction Temperature and 25°C ambient temperature):

Pv= (Tj – Ta) /Rthges = 58W or. 234W for 4 Units. For safety reasons the maximum power dissipation will be limited to 200W .

3.2 Generation of the control voltage for the constant current sink

Another requirement for this electronic load was that it should be able to handle AC current. In this case, the waveform of the control voltage must follow the waveform of the input voltage. To make this current controllable, the (divided) input voltage must be connected to X1 of an analog multiplier (see picture below). The shunt resistor is 0,1Ohm, so that a control voltage of 0,25V is necessary for a current of 2,5A. The (DC) control voltage is connected to Y1 of the multiplier. The output of the multiplier is directly connected to the input of all constant current sink units.

The multiplier used is an AD633:



X2 and Y2 are connected to ground, Z is connected to a negative voltage of 0...-100mV. The output voltage on W is:

W= (X1 * Y1)/10 +Z

Voltage on Z must be adjusted so that there is a current of 0mA at a destination current of 0mA .

The control voltage on Y1 will be calculated by the microcontroller and transferred to the 12bit D/A- cnverter IC6 . IC6 is operating at a reference voltage of 2,5V . This is also the maximum output voltage VOUT.

Datasheet:

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3. 3 Measuring input voltage

Measuring of the input voltage will be done by the microcontroller. To cover the whole range of 0 to 400V with sufficient resolution, two voltage ranges are used which are also controlled by the microcontroller (Signal Gain).

First, Q2 is on, division factor is R20/(R28||R21) = 1:0,015

This voltage is drirectly connected to the multiplier, for measuring the input voltage, this value is again divided by R22 and R23 by 10 and then multiplied by IC3 B with 3,26.

When the input voltage falls below 50V, Q2 is off and the input voltage will be divided by R20/R21 = 1 : 0,17.

At an input voltage of 50V the voltage on the output of IC3B is 2,77V (=Bitvalue 567 @10bit A/D conversion). The resolution is 100mV.

At an input voltage of 400V the voltage on the output of IC3B 1,956V (Bitvalue400). The resolution is 1V.

Depending on the input voltage (AC/DC) either the DC voltage or the rectified AC voltage will be measured. Calculation of the True RMS voltage will be done in Software (see Firmware).

Calibration of the voltage display will also be done in firmware via USB- interface (see Firmware and calibration).

3.4 Current measurement

The load current of all four channels will be added by IC7A and IC7B. At 2,5Arms the output amplitude is 1V. This value will be multiplied by 4,3 with IC7D, this makes a voltage of 0,43V/A (Bitvalue 88/A), resolution is 11mA.

Depending on the input voltage (AC/DC) either the DC current or the rectified AC current will be measured. Calculation of the True RMS current will be done in Software (see Firmware).

With R65, the offset of the current display must be adjusted, so that the display shows 0mA at a load current of 0mA .

Calibration of the current display will also be done in firmware via USB- interface (see Firmware and calibration).

Remark!

At 10A there is a voltage of 4,3V at the ADC Input. For AC currents, the peak value is then 14A, which makes 6,02V. This value is too high for current measurement, therefore the RMS value for AC current must not exceed 7A !

3.5 Microcontroller

An ATMega 32 microcontroller will be used for controlling this electronic load. The controller is operating at 16MHz with an external crystal and can be programmed via SV1 on the SPI interface. During programming, the controller must be supplied with the internal power supply.

Portdesignation:

Portpin	Connection	Remark
PA0 (ADC0)	Actual voltage	
PA1(ADC1)	Actual current	
PA2(ADC2)	Heatsink temperature	measured with NTC resistor B57045K0102K000, see Firmware
PA3(ADC3)	Potentiometer current destination value	
PB7(SCK)	SPI Clock	
PB6(MISO)	SPI Miso	only for programming
PB5(MOSI)	SPI Mosi	
PB4	CS_DAC	Chip select forD/A Converter
PB3	LDAC	Signal LDAC for A/D Converter
PB2	LDAC	Signal LDAC for Display
PB1	Switch Load On/Off	
PB0	Switch R-const/I-const	
PC7	Gain	switch gain range: 0= 0..50V 1= 50...400V
PC5	RelayLoad On/Off	1=On
PC4	Fan On/Off	1=On
PD7	LED Overvoltage	
PD6	LED Overcurrent	
PD5	LED SOA	
PD4	LED Overtemp	
PD3	LED Remote	
PD1	TXD	
PD0	RXD	

3.6 Display

For displaying voltage and current, 7 - segment- Displays are used, which are controlled by a MAX7219 Display Controller. The schematics was taken from the „Elektor Halbleiterheft 2009“ with a modified layout (distance between D3 and D4 enlarged).

Elektor article:

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Modified schematics and layout:

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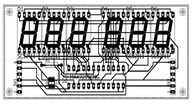

3.7 USB interface

The communication with a PC will be handled by a USB/Uart converter M2. To ensure galvanic isolation, the connection to the microcontroller is done with optocouplers OK3 and OK4.

Datasheet USB/Uart converter:

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3.8 Power supply

This electronic load needs +5V for the microcontroller and +12V and -12V for the analog part of the circtuitry. For that, two printtransformers are located on the PCB. Regulation of the voltages will be done by standard- voltage regulators (78XX and 79XX) .

4. Firmware

The firmware for the ATMega32 was written with BASCOM AVR.

The firmware uses 2 timer, the general purpose timer0, which generates an interrupt every 1ms and timer 2, which generates also an interrupt every 1ms and is necessary for calculating true RMS voltage and current. Timer 2 must be used necessarily, because sampling of voltage and current values must be done in exactly defined interval (see also 4.1). Interrupt-Priorities cannot be changed in the ATMega, therefore the interrupt with the highest priority must be used:

Table 18. Reset and Interrupt Vectors

Vector No.	Program Address[2]	Source	Interrupt Definition
1	$000[1]	RESET	External Pin, Power-on Reset, Brown-out Reset, Watchdog Reset, and JTAG AVR Reset
2	$002	INT0	External Interrupt Request 0
3	$004	INT1	External Interrupt Request 1
4	$006	INT2	External Interrupt Request 2
5	$008	TIMER2 COMP	Timer/Counter2 Compare Match
6	$00A	TIMER2 OVF	Timer/Counter2 Overflow
7	$00C	TIMER1 CAPT	Timer/Counter1 Capture Event
8	$00E	TIMER1 COMPA	Timer/Counter1 Compare Match A
9	$010	TIMER1 COMPB	Timer/Counter1 Compare Match B
10	$012	TIMER1 OVF	Timer/Counter1 Overflow
11	$014	TIMER0 COMP	Timer/Counter0 Compare Match
12	$016	TIMER0 OVF	Timer/Counter0 Overflow
13	$018	SPI, STC	Serial Transfer Complete
14	$01A	USART, RXC	USART, Rx Complete
15	$01C	USART, UDRE	USART Data Register Empty
16	$01E	USART, TXC	USART, Tx Complete
17	$020	ADC	ADC Conversion Complete
18	$022	EE_RDY	EEPROM Ready
19	$024	ANA_COMP	Analog Comparator
20	$026	TWI	Two-wire Serial Interface
21	$028	SPM_RDY	Store Program Memory Ready

The lower the vector, the higher the interrupt priority. INT0..INT2 are not used here ==> Timer2 has the highest Priority!

4.1 Measuring true RMS of voltage und current

The true RMS values are calculated as follows:

$$U_{\text{eff}} = \sqrt{\frac{1}{T}\int_{t_0}^{t_0+T} u^2\,\mathrm{d}t} = \sqrt{\overline{u^2}}$$

or for constant interval:

$$U_{\mathrm{eff}} \approx \sqrt{\frac{1}{n}\sum_{i=1}^{n} u_i^2} = \sqrt{\frac{1}{n}\left(u_1^2 + u_2^2 + u_3^2 + \cdots + u_n^2\right)}$$

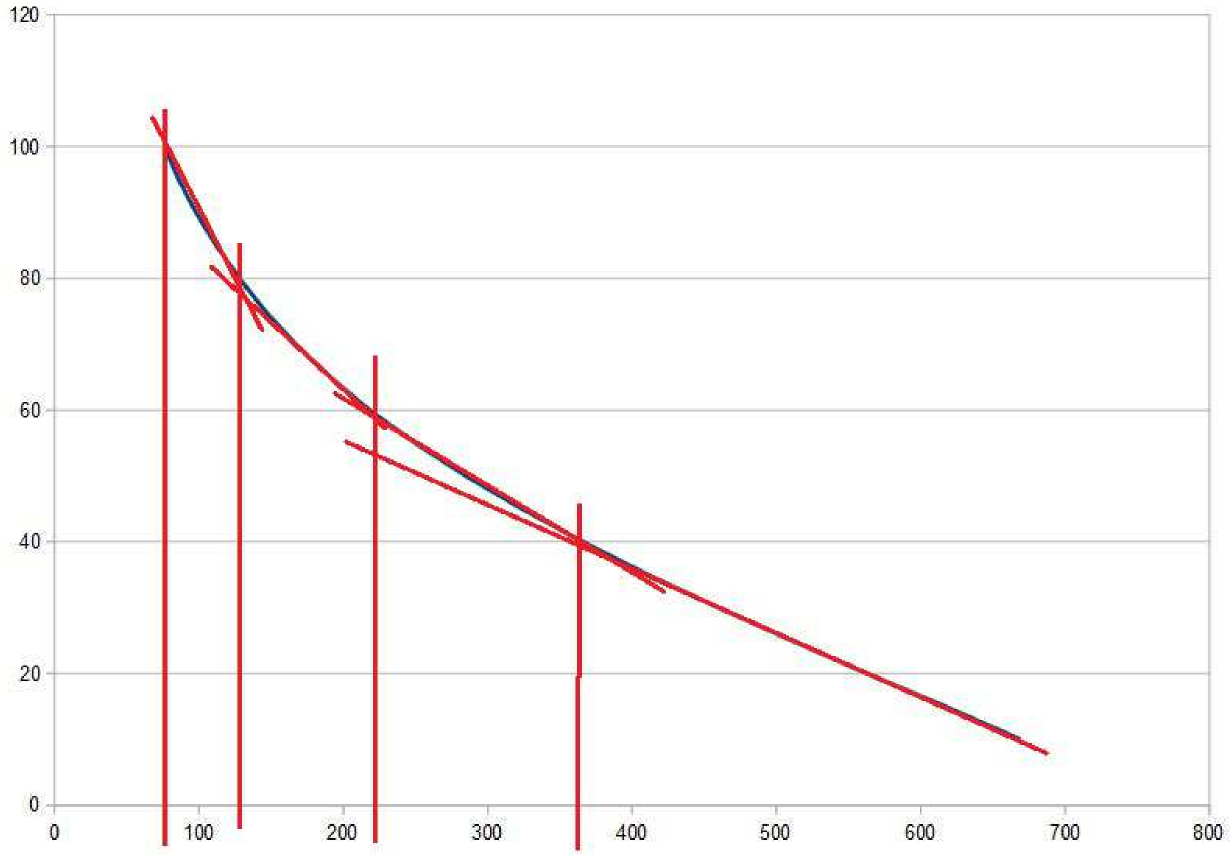
The constant interval is 1ms, after 20ms the true RMS value will be calculated. Therefore AC voltages and currents up to 100Hz can be measured.

In Timer2 Interrupt, u_i and i_i are sampled at the A/D inputs and stored in an Array.

After 20ms „Measure_completed" will be set to 1 and the true RMS value will be calculated. For that, the stored values are squared and added. The result will be divided by 20 and then the squareroot will be calculated. For „smoothing" of the true RMS values, an average of 10 consecutive measurements will be calculated. So one measurement takes 200ms!

4.2 Measurement of the heatsink temperature

The heatsink temperature will be measured on ADC2 with an NTC resistor of 1kOhm (B57045K0102K000). The curve is not linear, therefore it will be divided into 4 lines:



Y-Axis = Temperature in °C dar, X- Axis= ADC values

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The first line is from 0 bis 40°C, the second from 40 bis 60°C, the third from 60 bis 80°C and the forth from 80 bis 100°C. The function „Calculate_heatsink_temp“ is responsible for the calculation of the temperature.

4.3 Controlling of 7-Segment Display

The modules MAX7219.bas“ and „MAX7219.inc“ are responsible for controlling of the Display.

In MAX7219_init the chip will be set to 6 Digits and decoding of Code B.

In the function „Max7219_display(byval Leftdigits As Single , Byval Rightdigits As Single)“ only the values for voltage(Leftdigits) and current (Rightdigits) must be entered. They will be then displayed: the current always in format X.XX , the voltage in format XX.X when <100V, otherwise in format XXX.

4.4 Controlling of 12bit D/A converter

The modules „MCP4921.bas“ and „MCP4921.inc“ are controlling the DAC via SPI interface. There is only one function: Dac_out(byval Dacval As Word , Byval Gain As Byte)

Dacval = 12 bit output value,

Gain = 1: Vout = 2,5V * Dacval/4096, resolution = 610uV

Gain = 2: Vout= 5V * Dacval/4096, resolution = 1,2mV

4.5 Calculation of the destination current

Calculation of destination current depends on the setting of the R-const/I-const switch:

In constant current mode (I-const), the current must be constant. In constant resistor mode (R-const), the resistor must be held constantly by variying the current to I= Vinput/R-const.

To achieve a constant current, the DAC value must be calculated depending on the destination current and the actual (RMS) voltage.

The equation for the current is $I = U_{ic5W} / 0{,}1 * 4$

$U_{ic5W} = (0{,}15 Uist * Udac)/10$ for Uist <50V

$U_{ic5W} = (0{,}015 Uist * Udac)/10$ for Uist >50V

Conversion for the necessary DAC voltage:

Udac = 1,67 * 10^{-3} * (Isoll[mA]/Uist) for Uist < 50V

Udac = 1,67 * 10^{-2} * (Isoll[mA]/Uist) for Uist > 50V

Dac_value = (Udac * 4096) /2,5

This is performed in function „Set_dest_current“:

At an input voltage below 2V the destination current = „0“.

4.5.1 Current regulation

Because of component tolerances, destination current and actual current can be different. Therefore, in the function „Set_dest_current“ a regulation function with I- characteristic is implemented:

The function will be called every 200ms after calculation of the (RMS) voltage and current values. If the destination current is greater than the actual current, the control value will be increased by 5mA. If the destination current is less than the actual current, the control value will be decreased by 5mA.

4.6 Main program loop

In the main program loop the error conditions will be checked first:

- Voltage >400V: LED >U on, load relay off , switch on again if voltage <400V
- Current: check one second after changing destination current if actual value < destination current + 100mA. If not: LED >I on, load relay off. Load relay will be switched on again after switching load On/Off off and on.
- Power: If voltage * destination current >200W then LED >SOA on, Load relay off. Load relay will be switched on again after switching load On/Off off and on (and power <200W)
- Temperature: At a heatsink temperature of 40°C the fan will be switched on. It will also switched on when power exceeds 100W. When the temperaure falls below 35°C and the power is less than 100W for more than 10s, the fan will be switched off. When the heatsink temperature exceeds 80°C , the LED >T will be switched on and the load relay will be switched off. When the heatsink temperature falls below 75°C, the load relay will be switched on again.

This will be done every 500ms.Also the voltage and current values in the display will be updated every 500ms.

If there is no remote control (Remote=0), then the switches Load On/Off and I-const/R-const will be read as well as the potentiometer value for the destination current.

4.6.1 Setting destination current /resistor with potentiometer

After switching on the electronic load, the destination current range is 1A when set to I-const mode. After switching the load switch to „Off“, the destination current range will be set according to the input voltage: current range is 10A if the input voltage is less than 20V, otherwise destination current range is 200W/actual voltage.

In R-const mode, the minimum resistor is also calculated when the load switch is set to „Off“. It is Input voltage/destination current range. The maximum adjustable resistor value is minimum resistor * 100(Ohms).

Remark

When set to I-const mode, the left stop of the potentiometer meets the minimum current, right stop the maximum current. To avoid overcurrent when switching from I-const mode to R-const mode, the left stop value of the potentiometer corresponds to the maximum resistor value.

4.6.2 Serial interface

The serial interface of the microcontrollers works with 19200Baud, 8Databits, 1Stopbit, no Parity. The main program loop checks if there are characters received. The following protocol is implemented:

<Command>[<Parameter(s)>]<CR><LF>

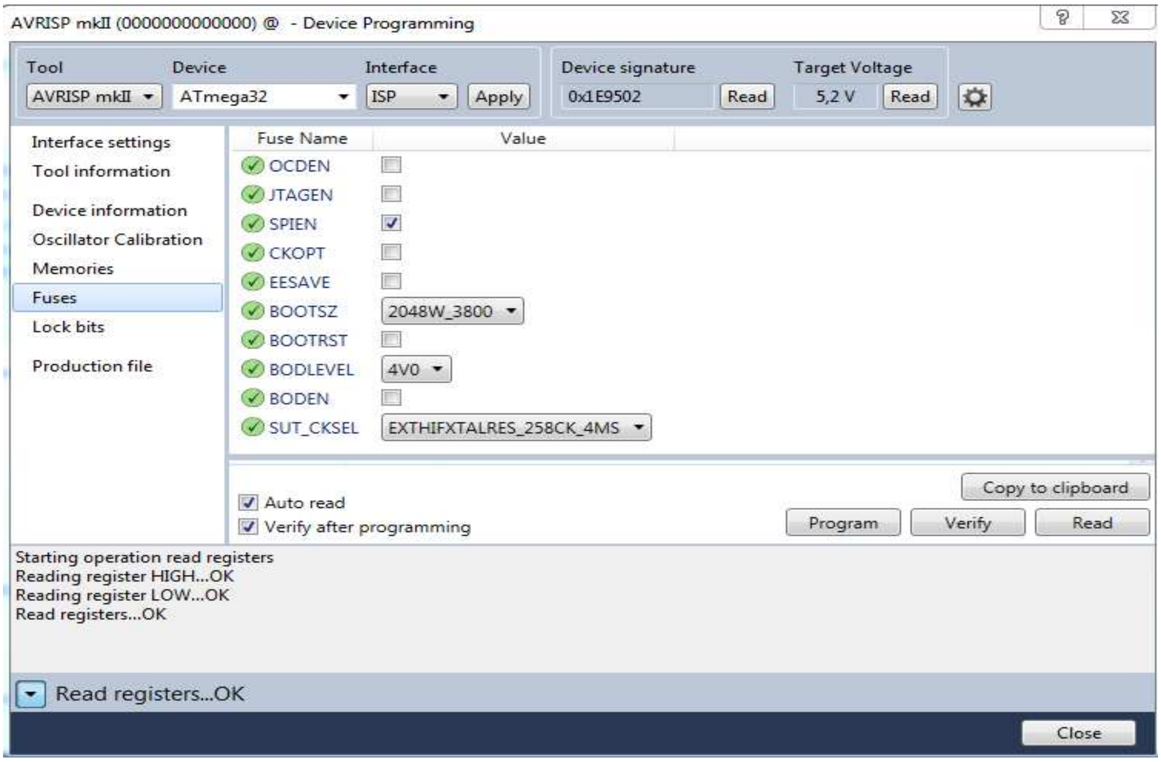
Following commands are implemented:

Command	Parameter	Response	Meaning
A	-	AAC DC Load V1.0	Identification string
B	0/1	-	1= Remote On, 0= Remote Off
C	-	CVrms (V)	Voltage (RMS)
D	-	DmArms	Actual current (RMS)
E	-	EmA	Destination current
F	0...9999	-	Set destination current in mA Only if Remote on
G	-	0,1,2,4,8	Error: No_error = 0 Overvoltage = 1 Overcurrent = 2 Overtemp = 4 Soa_error = 8
H	0..99	-	Heatsink temperature in °C
I	0/1	-	Switch Regulation mode 0=I-const, 1= R-const Only if Remote on
J	0/1	-	Load relay On/Off On only if no error Reset error conditions if off Only if Remote on
K	-	0/1	Condition of load relay
L	-	0/1	Condition of regulation mode 1= I const, 2= R const
M	-	0...200W	Power in W
N	-	0..9999	Destination current range in mA
O	-	0/1	Condition Remote 0=Off, 1=On

Command	Parameter	Response	Meaning
P	0..100	-	Set calibration value for current
R	-	0..100	Request calibration value for current
S	0..10	-	Set calibration value for voltage
T	-	0..100	Request calibration value for voltage
U	-	U u1,u2,.....u20 i1,i2,...i20	Momentarily values for voltage/current (only for testing purpose)
V	-	VXXXXX	send minimum resistor in Ohms
W	-	WXXXXX	send destination resistor range in Ohms
X	XXXXX	-	set destination resistor in Ohms only if remote on
Y	-	YXXXXX	send destination resistor in Ohms

4.7 Programming

The microcontroller can be programmed via SPI interface with any suitable programmer (avrISP mkII for example). The fuse bits must be set as shown here:



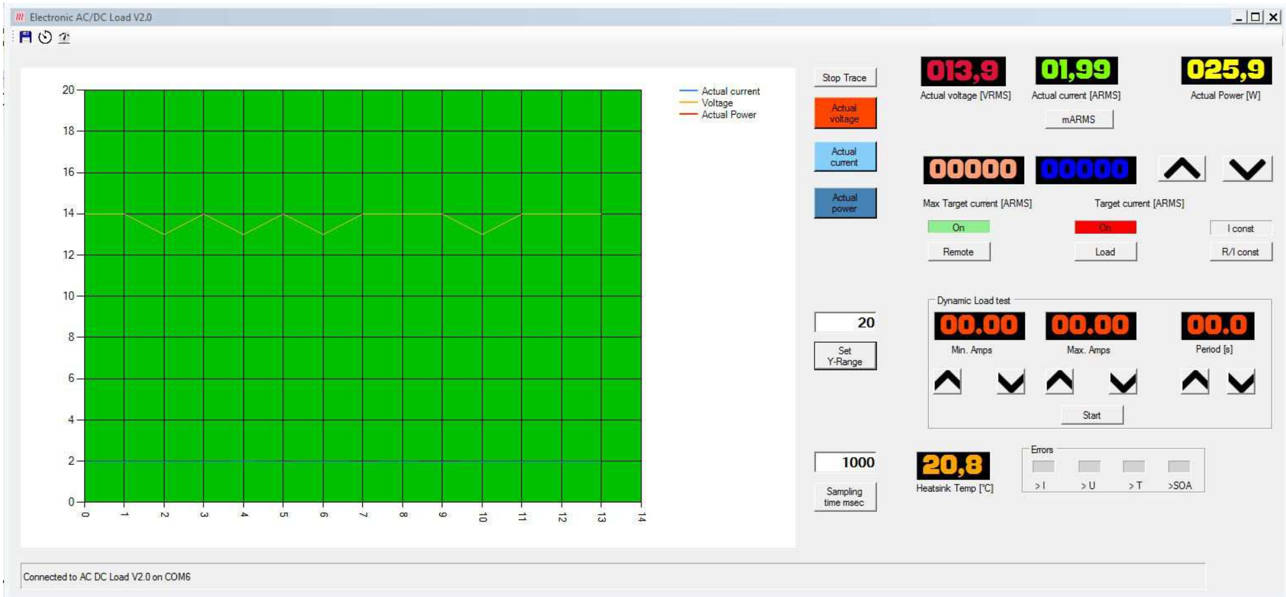
Important safety information!

When the electronic load is connected to the programmer, there is no galvanic isolation from the input terminals. So be sure that the programmer is disconnected when the input terminals are connected to mains or when the negative input terminal is not connected to ground!

5. Control software

The control software for the elctronic load was written in in Visual Basic for Visual Studio 2015. The executable file is „Electronic_ACDC_Load.exe“.

After invoking the program, the following window will be displayed:



The program tries to connect to the load by scanning all available COM ports and requesting the ID string every second for a maximum of 3 times. When the program receives the IDString, it will be displayed in the status line together with the COM Port.

Then all data for voltage, currents, power and heatsink temperature will be requested every second and will be displayed in the appropriate display.

For setting the destination current/resistor, the Button „Remote“ must be set to „On“ . Then the load relay (Load On/Off) and the R-const/I-const- mode can be set by the program.

With the button „mARMS“ the display can be switched from „A“ to „mA“ and vice versa (only in I-const mode).

With the Up/Down buttons, the destination current can be increased/decreased (Increasing is possible up to „Max destination current“).Change steps starts with 10mA and increases to 100mA and finally to 1A steps. After release of the keys, the change step is again 10mA (only in I-const mode).

In R-const mode, the destination resistor can be increased/decreased (Increasing is possible without limitation), decreasing is possible down to the minimum resistor .Change steps starts with 1 Ohm and increases to 10 Ohm and finally to 100 Ohm steps. After release of the keys, the change step is again 1 Ohm.

5.1 Dynamic Load Test

Here you can set two destination currents (I-const mode), or two destination resistors (R-const mode) to test power supplies for example.With „Period“ you can set the interval in seconds. After pressing „Start“ the two destination currents/resistors are switched from the first value to the second and so on until you press „Stop“. Then the current will be set to „0“ in I-const mode or set to maximum resistor in R-const mode

5.2 Displaying of voltage, current and power traces

Aftter pressing the key „Start Trace“ , the previously activated traces „Actual Current“, „Actual Voltage“,and „Actual Power“ can be displayed. The recording speed can be set with the key „Sampling time msec“. A maximum of 60 points will be displayed, afterwards the traces will be cleared and the recording starts again. The same occurs if you change any target. Resolution of the vertical axis can be changed with „Set Y-Range“ .

5.3 Log file

By clicking on the diskette symbol, a log file can be opened for long time recording of the data. The data is stored in .csv format, so you can continue processing the data in Excel or OpenOffice.

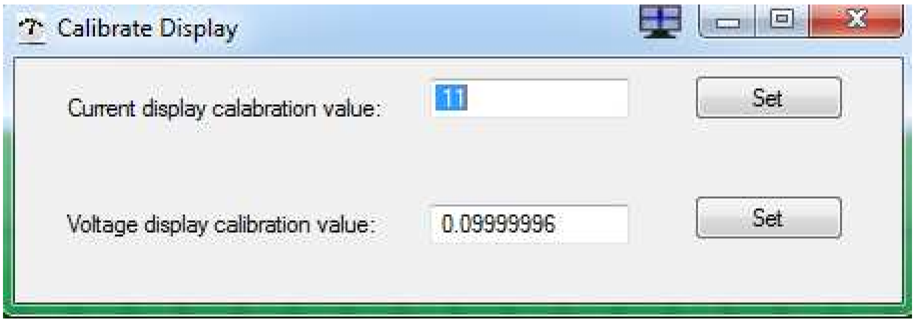
Data are stored as follows

	A	B	C	D	E	F	G
1	Time	Actual voltage[V]	Actual current[A]	Actual power[W]	Destination current[A]	Destination resistance[Ohms]	Heatsink temp. [├≋C]
2	15:59:32	17	0.02034683	0.3458962	0	178.6093	22.6
3	15:59:33	17	0.019387	0.329579	0	178.6093	22.6
4	15:59:34	17	0.02057335	0.3497469	0	178.6093	22.6
5	15:59:35	17	0.02063568	0.3508066	0	178.6093	22.6
6	15:59:36	17	0.01904882	0.32383	0	178.6093	22.6
7	15:59:37	17	0.01883507	0.3201962	0	50	22.6
8	15:59:38	17	0.0187488	0.3187296	0.03	30	22.6
9	15:59:39	17	0.01797906	0.3056439	0.3536206	50	22.6
10	15:59:40	16	0.4549854	7.279766	0.559552	30	22.6
11	15:59:41	16	0.249405	3.99048	0.337767	50	22.6
12	15:59:42	16	0.4835536	7.736858	0.5590218	30	22.6
13	15:59:43	16	0.2791201	4.465922	0.3370324	50	22.6
14	15:59:44	16	0.5090477	8.144764	0.5585514	30	22.6
15	15:59:45	16	0.311267	4.980272	0.337224	50	22.6
16	15:59:46	16	0.5392121	8.627394	0.5588668	30	22.6
17	15:59:47	16	0.3263148	5.221036	0.3373624	50	22.6
18	15:59:48	16	0.5571177	8.913883	0.5577423	30	22.6
19	15:59:49	16	0.3521191	5.633906	0.3366041	50	22.6
20	15:59:50	16	0.5581734	8.930774	0.5572993	30	22.6
21	15:59:51	17	0.3369544	5.39127	0.3372261	50	22.6

By clicking on the timer symbol, the Sampling time for the Logfile can be changed (Default = 1 Second). An opened Logfile can be closed by clicking the diskette symbol again.

5.4 Calibration

By clicking on the instrument symbol, display of the electronic load can be calibrated:



You need calibrated test equipment for measuring voltage and current. By changing the values for „Current display calibration value" and „Voltage display calibration value" you can adjust the display values to the readings of the calibrated instruments.

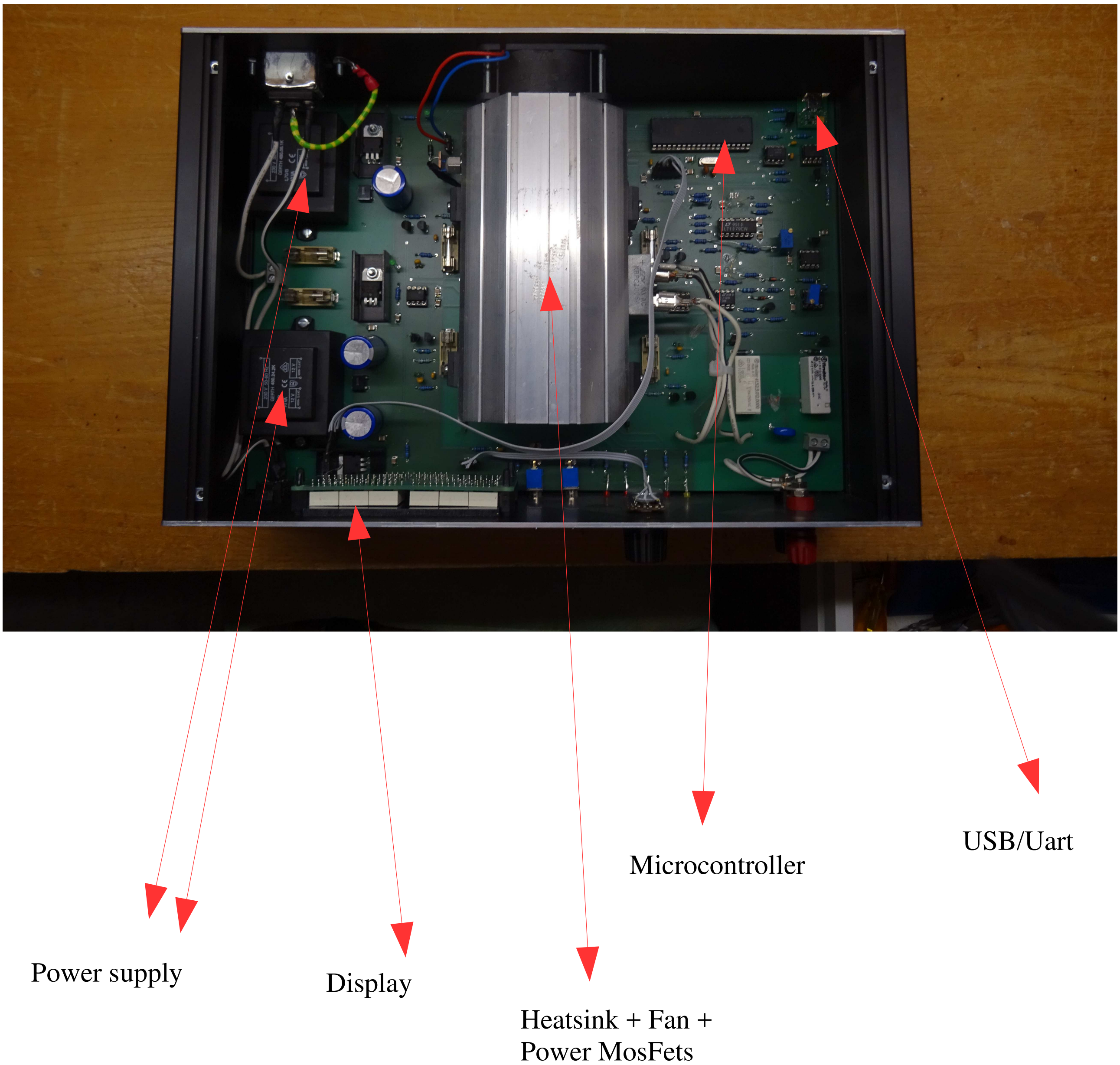
The values are stored in EEPROM of the microcontroller.

5.5 Communication error

The program requests data from the electronic load every second. If there is no answer within 3 seconds, the program will be terminated with the following error message:

6. Construction

The PCB was built in a case:



The power mosfets are mounted on the heatsink with TOP3 isolating foiles and thermally conductive paste.

The heatsink itself is mounted on the PCB wit 2mm plastic distance holders (M3).

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